

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (90 minutes) April 6th, 2013

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. Let X_1, X_2 be a random sample of size 2 from a gamma distribution with parameters $\alpha = 3$ and θ , that is

$$f_X(x; \theta) = \frac{e^{-x/\theta} X^2}{\Gamma(3) \theta^3} x > 0.$$

- a) Find a sufficient statistic.
- b) Is $\log e(x_1, x_2)$ sufficient?
- c) Find the MLE of e^θ .

2. Let X be a random sample of size 1 having an exponential density with parameter θ . That is

$$f_X(x; \theta) = \frac{1}{\theta} e^{-x/\theta}, x > 0, \theta > 0.$$

Find the most powerful test of size e^{-1} for $H_0: \theta = 1$ vs $H_1: \theta = 3$.

3. Consider a random permutation of $1, 2, \dots, 2n$ (each permutation is equally likely). What is the probability of existing exactly k numbers i such that $2i - 1$ immediately precedes $2i$ when $n \rightarrow \infty$?

4. Let X and Y be independent random variables, each with distribution $U(0,1)$.

- a) What is the probability that at least one of the two is larger than 0.8?
- b) What is the probability that the sum $x+y$ is smaller than 1?